

Bishowjith Ghosh

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M.Sc. in Data Science

B.Sc. in Chemistry

| US Citizen, Pflugerville, Texas

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| Birchwood University, Florida, USA (2025)

| National University, Bangladesh (2013)

CAREER OBJECTIVE

A highly skilled and results-driven professional with over 10 years of hands-on experience in manufacturing and a strong foundation in data science. Combining deep expertise in production optimization, system configuration, and quality assurance with advanced proficiency in data analysis, machine learning, and business intelligence tools, including Tableau and MySQL. Currently holding a Master's degree in Data Science, I am seeking to leverage my diverse skill set in a dynamic role that enables me to contribute to data-driven decision-making, process optimization, and business success across multiple industries. I am committed to continuous professional growth and delivering actionable insights that drive organizational performance, efficiency, and innovation.

CORE COMPETENCIES - Manufacturing & IT Integration

Manufacturing Processes | Production Optimization | Quality Assurance | 6S Methodology | OSHA Standards 6s | Lockout/Tag Out | Clean Room Protocols | System Configuration | Robotic Tool Setup & Installation | Cable & Wiring Installation | Component Assembly | Sub-Assembly | Troubleshooting & Diagnostics | IOMS & MES Integration | Water Pipeline Testing | Electronics Inspection | Circuit Board Tray Installation | PCB Assembly & Testing | Mainframe Operations | Semiconductor Tools Setup & Maintenance | Production Work-flow Optimization | Lean Manufacturing | Assembly Line Balancing | Team Collaboration | Workforce Training & Development | Safety Protocols & Compliance | Time Management & Efficiency | Shipping & Packaging | Order Verification & Dispatch

Data Science & Analysis

Data Analysis: Data Cleaning, Data Mining, Exploratory Data Analysis (EDA)

Data Integration: Python-MySQL Integration, Data Extraction, SQL Queries

Data Visualization: Tableau (Interactive Dashboards, Storytelling, Data Reporting)

Programming: Python (Pandas, NumPy, Seaborn), MySQL, Scikit-learn, Flask (for ML deployment), GitHub, Jupyter Notebook

Machine Learning: Supervised & Unsupervised Learning, Regression Models, Predictive Analytics

Data Modeling & Validation: Time Series Analysis, Data Transformation, Integrity Management

Technical Tools: Microsoft Office Suite

Soft Skills: Problem-Solving, Critical Thinking, Effective Communication, Team Collaboration

Certification & Professional Development

Information Technology:

Data Analyst Professional Certificate of Completion, Taskimpetus | Aug 2024

Software Testing and Automation Course, NextTech ITC, Irving, TX | Mar 2021

CompTIA IT Fundamentals, austinfree.net, Austin, Texas | Mar 2018

CompTIA A+ Complete Study Guide (Course & Labs), The Goodwill Academy | May 2018

Diploma in Computer Science & ICT, SIT Foundation, Bangladesh | Dec 2010

Manufacturing:

Lock out/Tag out, Crane Hoist & lifting, Safety, IOMS for manufacturing, Electrical Safety, 6s and more.

Work Experience:

Manufacturing Technician (Production Data & Process Improvement)

Applied Materials, Austin, TX | Sep 2021 – Present

Currently in Building 36 (ETCH Department)

Previously worked in Building 34 (CSMG Department) and Building 32 (Endura Department).

Semiconductor Industry Experience:

- **System & Tool Assembly:** Building systems/tools from scratch, including assembly and sub-assembly of components, following strict clean room protocols to ensure system cleanliness and environmental integrity.
- **Cabling & Wiring:** Installing and routing cables, wiring for servers, UPS systems, Moxa, and robotic tools, ensuring proper alignment and connection for optimal system performance and stability.
- **Etch/Endura/Csmg Department Support:** Assembling and cabling Centris, AP, Falcon, Mainframe systems, ensuring the tools were properly set up to optimize semiconductor production processes in the department.
- **Installation & Leak Testing:** Installing electrical cables, copper, steel pipelines, and poly lines in the Mainframe Endura LPG area, conducting leak tests, building gas boxes, and ensuring system integrity and safety compliance.
- **Quality Control:** Performing water pipeline testing and inspecting systems for compliance with operational requirements, ensuring readiness and tool performance.
- **Troubleshooting & Maintenance:** Repairing, installing, and maintaining electronic systems in the CSMG Fise Build Area, troubleshooting PCB boards to ensure steady production flow.
- **Packing & Deployment:** Assisting in packing and shipping tools after installation and QA checks, ensuring optimal tool readiness for deployment.
- **Team Training:** Training new personnel, emphasizing safety protocols, and collaborated with the team to meet and exceed quarterly production targets.

IT & Data Integration in Manufacturing:

- **Data Analysis:** Managing and analyzing production data for semiconductor tools using Python for data cleaning, preprocessing, and improving performance accuracy.
- **Database Management:** Utilizing MySQL to query and manage large datasets from production systems, optimizing data retrieval for performance analysis.
- **Real-Time Monitoring:** Integrating data from IOMS and MES for real-time monitoring of tool performance, identifying areas for process improvement and enhancing operational efficiency.
- **Data Visualization:** Developing interactive Tableau dashboards for tracking production metrics, tool performance, and predictive maintenance models, driving data-driven decision-making.
- **Predictive Maintenance:** Implementing predictive maintenance models using real-time data, reducing downtime by 25% and improving overall equipment effectiveness (OEE).

Quality Assurance & Collaboration with IT Teams:

- **Quality Standards Compliance:** Ensuring strict adherence to quality standards by inspecting and testing equipment, collaborating with IT professionals to integrate data systems that align manufacturing goals.
- **System Integration:** Working closely with IT teams to ensure seamless integration of Manufacturing Execution Systems (MES) and Integrated Operations Management Systems (IOMS), streamlining production work-flows and tool performance.
- **Cross-Functional Collaboration:** Regularly communicating with IT teams to troubleshoot performance issues and optimize systems, ensuring alignment between manufacturing and IT for maximum efficiency.

Cross-Disciplinary Expertise:

Hands-On Manufacturing & IT Integration: Combining expertise in system assembly, cabling, and troubleshooting with data analysis, system integration, and tool performance monitoring to enhance semiconductor manufacturing processes.

Clean Room Technician: Luminex Corp, Austin, TX | Feb 2021 – June 2021

- Built and tested parts for COVID-19 testing kit assemblies, ensuring compliance with FDA medical device standards.
- Operated specialized equipment for the assembly, testing, and packaging of diagnostic kits.
- Assisted in the optimization of cleanroom processes to improve production efficiency while maintaining compliance with health standards.
- Supported troubleshooting and repairs of assembly machines to ensure minimal downtime and high production rates.
- Provided feedback on the manufacturing process, supporting investigations and process changes.
- Maintained cleanroom protocols to uphold the highest standards of hygiene and contamination control.
- Assisted in editing procedures and records for the build and test process, ensuring accuracy and alignment with regulatory requirements.

Customer Service: Super Mart, Austin, TX | Aug 2020 – Dec 2020

- Provided exceptional customer service to a diverse clientele, addressing inquiries and resolving issues in a timely manner.
- Processed sales transactions accurately and efficiently using the point-of-sale (POS) system.
- Assisted customers in locating products, offering suggestions based on preferences, and ensuring customer satisfaction.
- Supported inventory management by restocking shelves, organizing products, and maintaining store cleanliness.
- Promoted store loyalty programs, effectively increasing membership enrollment.
- Participated in store merchandising to optimize product displays and enhance sales visibility.
- Maintained high levels of professionalism and courtesy in all customer interactions, contributing to a positive shopping experience.

Assembler II: AirBorn Inc, Georgetown, TX | Feb 2019 – July 2020

- Conducted quality assurance for cable assemblies, with a focus on wire harness connectors for military and aerospace applications.
- Rotated through production tasks, including assembling, inspecting, and testing materials to meet specifications.
- Created marking files using ROFIN laser machines for precise and accurate labeling of components.
- Installed and connected cables and tubing per detailed specifications to ensure operational integrity.
- Generated inspection reports and non-conformance documentation to ensure customer requirements were met.
- Participated in team meetings to discuss production updates, troubleshoot issues, and collaborate on process improvements.
- Ensured compliance with safety protocols and industry standards to maintain high-quality product output.

Assembler: X Desk, Pflugerville, TX | Mar 2018 – Jun 2018

- Assembled computer desks, keyboards, and chairs, following established safety policies.
- Used RF scanners for order picking and shipping, maintaining organization and cleanliness in the workspace.
- Loaded and unloaded items while ensuring quality and accuracy in product handling.
- Assembled products based on customer specifications and design blueprints to meet product requirements.
- Conducted quality checks on assembled products, ensuring they met standards before shipment.

Production Operator: Build A Sign, Austin, TX | Jan 2017 – Feb 2018

- Managed the production and assembly of plastic and magnetic signs using specialized machinery, ensuring quality and efficiency.
- Verified orders, boxed products, and prepared shipments while following safety protocols.
- Worked across multiple departments (canvas, small flex, T-shirt) using various tools to enhance productivity and meet deadlines.
- Performed routine maintenance and troubleshooting to prevent delays and ensure work-flow.
- Operated large-scale machines, including the Zund cutter, to streamline sign production.
- Processed shipping information and ensured accurate packaging and dispatch of orders.
- Participated in the continuous improvement of assembly line processes to enhance overall efficiency and reduce downtime.

Customer Service Cashier: Big Daddy's, Johnson City, TX | Oct 2016 – Jan 2017

- Operated cash register and balanced transactions.
- Provided on-time, friendly customer service and greeted customers appropriately.
- Monitored checkout stations to ensure adequate cash availability.
- Supervised work activities of sales representatives and processed customer returns.
- Assisted customers with pumping gas or pumped gas for them when required.
- Maintained clean and orderly areas around gas pumps and the convenience store.
- Ordered and received stock, ensuring items were displayed on shelves.
- Labeled items with correct prices and managed price modifications.
- Took cash and credit card payments and returned change.
- Practiced suggestive selling to encourage additional purchases.
- Received fuel trucks and supervised refilling of fuel pumps.

Customer Service Cashier: Valero Corner Store, Austin, TX | Jun 2016 – Oct 2016

- Operated cash register and balanced transactions.
- Provided friendly customer service and greeted customers appropriately.
- Counted money in cash drawers at the beginning of shifts to ensure adequate change.
- Supervised sales representatives and processed customer returns.
- Monitored checkout stations for adequate cash availability.
- Verified the age of customers purchasing alcohol or cigarettes.
- Assisted customers with pumping gas or pumped gas for them when required.
- Maintained clean and orderly checkout areas, including general cleaning duties.
- Ordered and received stock, ensuring items were displayed on shelves.
- Assisted with receipts, refunds, or change due to customers.
- Labeled items with correct prices and managed price modifications.
- Ensured the cleanliness of the gas pump and convenience store areas.
- Worked as a gas attendant when necessary.

Customer Service Cashier | Randall's, Austin, TX | Apr 2016 – Sep 2016

- Greeted customers and verified the age of those purchasing alcohol or cigarettes.
- Gained experience with cash registers, credit card reader, desktop computers.
- Processed receipts and bills for customers and developed positive customer relationships.
- Organized racks and shelves to maintain store appearance.
- Kept both departments clean, including sanitizing meat-cutting equipment.
- Set up the seafood case when the seafood manager was unavailable.
- Ensured the cleanliness of checkout areas, including general cleaning duties.
- Managed incoming shipments, inventory, and freshness before stocking.

Restaurant Crew | Wendy's, Austin, TX | Jan 2016 – Apr 2016

- Welcomed customers and provided menus, suggesting items based on preferences.
- Informed customers of special or new deals.
- Took orders and ensured accuracy by repeating them to customers.
- Processed payments and relayed orders to the kitchen.
- Prepared orders, including fountain drinks, shakes, and ice cream.
- Assembled and packed orders securely, adding condiments as needed.
- Managed food preparation at fry stations and grills.
- Maintained cleanliness of preparation counters and equipment.

Web Designer: Uding Inc., Barisal, Bangladesh | Feb 2009 – Nov 2015

- Designed and developed websites, working closely with clients to understand their needs and ensure satisfaction.
- Managed the full web development life-cycle, including planning, designing, coding, and maintaining websites.
- Optimized websites for usability, performance, and SEO, ensuring maximum reach and engagement.
- Collaborated with graphic designers and content creators to produce visually appealing and functional websites.
- Implemented and maintained content management systems (CMS) to allow clients to easily update and manage their own content.
- Provided technical support and troubleshooting for website-related issues to clients.
- Developed custom features and functionality using HTML, CSS, JavaScript, and PHP to meet specific business needs.

Data Analysis Projects | 2024 - 2025

Sales Performance and Trends Analysis

- Developed a comprehensive Tableau dashboard to showcase sales performance, customer segmentation, and product analysis.
- Conducted data cleaning using Python (Pandas) and integrated MySQL for querying large datasets.
- Created visualizations for sales trends, product category sales, and geographical insights.

Bank Customer Data Analysis

- Built an interactive Tableau dashboard focusing on customer demographics and regional distribution.
- Utilized Python for data preprocessing and MySQL for efficient customer data management.
- Highlighted customer segmentation based on spending patterns and job classifications.

Retail Sales Insights

- Designed a Tableau dashboard to provide insights into sales performance, product analysis, and customer segmentation.
- Performed data wrangling in Python and utilized MySQL for complex data queries.
- Visualized sales, shipping performance, and regional analysis within Tableau.

Customer Data Insights

- Developed a Tableau dashboard to analyze customer behavior, sales performance, profit margins.
- Cleaned and transformed raw e-commerce data using Python, with data extraction from MySQL.
- Provided a detailed view of category-wise sales and yearly sales trends.